

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 5: Cluster Analysis

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
- **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
- **M3** DBSCAN point classification *practice*

M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

QUESTION AS PRINTED 2078 Bhadra · Q8 · [2+6]

Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

QUESTION AS PRINTED 2074 Ashwin · Q7 · [2+8]

Write K-means algorithm and find clusters for following data set.

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

(Take $K = 2$)

The same 6 instances, only the column names differ.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow$ new centroid $(\frac{1+2.5}{2}, \frac{2+1}{2}) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow$ new centroid $(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}) = (4, 2)$

Iteration 2

Point	d to $(1.75, 1.5)$	d to $(4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged $\rightarrow$ converged.**

Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity — worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, \dots, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**. Same data, same K , **different answer** — this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. verified/added

Problem 1.3 4+6

QUESTION AS PRINTED 2075 Ashwin · Q7 · [4+6]

Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering.

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

- Seeds: $c_1 = (1, 2)$ (clearly isolated), $c_2 = (4, 6)$.
- **Iteration 1:** $(1, 2) \rightarrow C1$. All of $(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)$ are nearer $(4, 6) \rightarrow C2$.
 - Eg $(2.5, 4.5)$: d to $c_1 = \sqrt{1.5^2 + 2.5^2} = 2.92$; to $c_2 = \sqrt{1.5^2 + 1.5^2} = 2.12 \rightarrow C2$.
 - Eg $(3.5, 4)$: d to $c_1 = \sqrt{2.5^2 + 2^2} = 3.20$; to $c_2 = \sqrt{0.5^2 + 2^2} = 2.06 \rightarrow C2$.
- New centroids: $C1 = (1, 2)$; $C2 = (\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5}) = (3.4, 5.2)$
- **Iteration 2:** every C2 member stays nearer $(3.4, 5.2)$ than $(1, 2) \rightarrow$ unchanged $\rightarrow$ **converged**.

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid $(3.4, 5.2)$.

- **Note:** $(1, 2)$ is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Problem 1.4 10

QUESTION AS PRINTED 2071 Shrawan · Q8 · [10]

Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset.

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

The paper prints no K ; take $K = 2$, the value its sister questions use.

- Seeds $c_1 = (1, 1)$, $c_2 = (5, 7)$.
- **Iteration 1:** $(1, 1)$ and $(1.5, 2) \rightarrow C1$; the rest $\rightarrow C2$.
 - $(3, 4)$: d to $c_1 = \sqrt{4 + 9} = 3.61$; to $c_2 = \sqrt{4 + 9} = 3.61$ – **an exact tie**. Break it toward $C2$ (state your rule).
- Centroids: $C1 = (1.25, 1.5)$; $C2 = \left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = (3.9, 5.1)$
- **Iteration 2:** assignments unchanged $\rightarrow$ converged.

Final answer

$C1 = \{(1, 1), (1.5, 2)\}$ centroid $(1.25, 1.5)$; $C2 = \{(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)\}$ centroid $(3.9, 5.1)$.

Problem 1.5 8

QUESTION AS PRINTED 2076 Chaitra · Q9 · [8]

Given the matrix X whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here K is chosen as 3. The center of the 3 clusters are initialized as red $(6.2, 3.2)$, green $(6.6, 3.7)$ and blue $(6.5, 3.0)$. Provide the final cluster centers and comment on the number of iterations required for the clusters to converge.

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red $(5.171, 3.171)$, green $(5.5, 4.2)$, blue $(6.45, 2.95)$

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red $(4.8, 3.05)$, green $(5.3, 4.0)$, blue $(6.2, 3.025)$

Iteration 3

- Every point stays where it was $\rightarrow$ **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.

- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2-6.6$), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Problem 1.6 – the defective question 5+3

QUESTION AS PRINTED 2081 Baishakh · Q4 (b) · [5+3]

Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.

(nothing follows on the paper – the next line is Q5)

The paper prints NO data table for this question – it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (\$5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points – it is the dataset this paper's sister questions use. verified/added

M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
 2. Find the **smallest** entry → merge those 2 clusters. **Record the merge height** – that value is the dendrogram bar.
 3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
 4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
- **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q4 (b) · [2+7]

What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points.

	p_1	p_2	p_3	p_4	p_5
p_1	0.00	0.24	0.22	0.37	0.34
p_2	0.24	0.00	0.15	0.20	0.14
p_3	0.22	0.15	0.00	0.15	0.28
p_4	0.37	0.20	0.15	0.00	0.29
p_5	0.34	0.14	0.28	0.29	0.00
p_6	0.23	0.25	0.11	0.22	0.39

The paper prints the full symmetric 6×6 ; the p_6 column is 0.23, 0.25, 0.11, 0.22, 0.39, 0.00.

Distances are given directly, so no coordinate work. The paper does not name a linkage — use **single link** and say so.

Merge sequence

Step	Merge	Height	Updated distances (MIN)
1	p_3, p_6	0.11	to p_1 0.22, p_2 0.15, p_4 0.15, p_5 0.28
2	p_2, p_5	0.14	to p_1 0.24, p_4 0.20, $\{p_3 p_6\}$ 0.15
3	$\{p_3 p_6\}, p_4$	0.15	to p_1 0.22, $\{p_2 p_5\}$ 0.15
4	$\{p_3 p_4 p_6\}, \{p_2 p_5\}$	0.15	to p_1 0.22
5	all, + p_1	0.22	—

Final answer

Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22.**

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

Problem 2.2 8

QUESTION AS PRINTED 2080 Baishakh · Q7 · [8]

Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance.

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Step 1: Euclidean distance matrix

$p_1(0.40, 0.53)$ $p_2(0.22, 0.38)$ $p_3(0.35, 0.32)$ $p_4(0.26, 0.19)$ $p_5(0.08, 0.41)$ $p_6(0.45, 0.30)$

	p_1	p_2	p_3	p_4	p_5
p_2	0.234				
p_3	0.216	0.143			
p_4	0.368	0.194	0.158		
p_5	0.342	0.143	0.285	0.284	
p_6	0.235	0.244	0.102	0.219	0.386

Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3p_6\}$ 0.386
3	$\{p_3p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2p_5\}$ 0.386
4	$p_1, \{p_2p_5\}$	0.342	to $\{p_3p_4p_6\}$ 0.386
5	the two remaining	0.386	—

Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.

Problem 2.3 4+4

QUESTION AS PRINTED 2076 Chaitra · Q10 · [4+4]

The table below is a distance matrix for six objects:

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.

The ideal question for showing you understand the difference — one matrix, both linkages.

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → **chaining**.

(b) Complete link (MAX)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** — a balanced split.

The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A-B-C-D chain and leaves E as a straggler, while complete link produces two clean balanced groups.

Single link chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

M3. DBSCAN point classification

Method

1. For each point, count how many points lie within ε (include the point itself).
2. Count $\geq \text{MinPts}$ → **core**.
3. Otherwise, within ε of a core point → **border**.
4. Otherwise → **noise**.
5. Join core points within ε of each other into clusters; attach the border points.

Practice

PROBLEM not a PYQ · written to fill the gap verified/added

Classify each of the points $A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(8, 8)$, $E(8, 9)$, $F(25, 25)$ as a **core**, **border** or **noise** point under DBSCAN with $\varepsilon = 2$ and $\text{MinPts} = 3$ (Euclidean distance, the point itself counted in its own neighbourhood). Then give the clusters DBSCAN produces.

No PYQ sets a DBSCAN numerical, but the algorithm is asked in 8 papers and a worked classification makes the definitions concrete. verified/added

Step 1: neighbourhood counts

(including the point itself)

Point	Points within $\varepsilon = 2$	Count	Type
$A(1, 1)$	A, B (1.00), C (1.00)	3	core
$B(1, 2)$	B, A (1.00), C (1.41)	3	core
$C(2, 1)$	C, A (1.00), B (1.41)	3	core
$D(8, 8)$	D, E (1.00)	2	border
$E(8, 9)$	E, D (1.00)	2	border
$F(25, 25)$	F only	1	noise

Step 2: form clusters

- A, B, C are all core and all within ε of each other → **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** → they are **not border points after all, they are noise**.
- F is isolated → **noise**.

Final answer

- **1 cluster**: $\{A, B, C\}$. **Noise**: D, E, F .
- **The trap**: D and E look like a cluster to the eye, but with $\text{MinPts} = 3$ a pair can never form one. **A border point must be within ε of an actual core point** — being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.